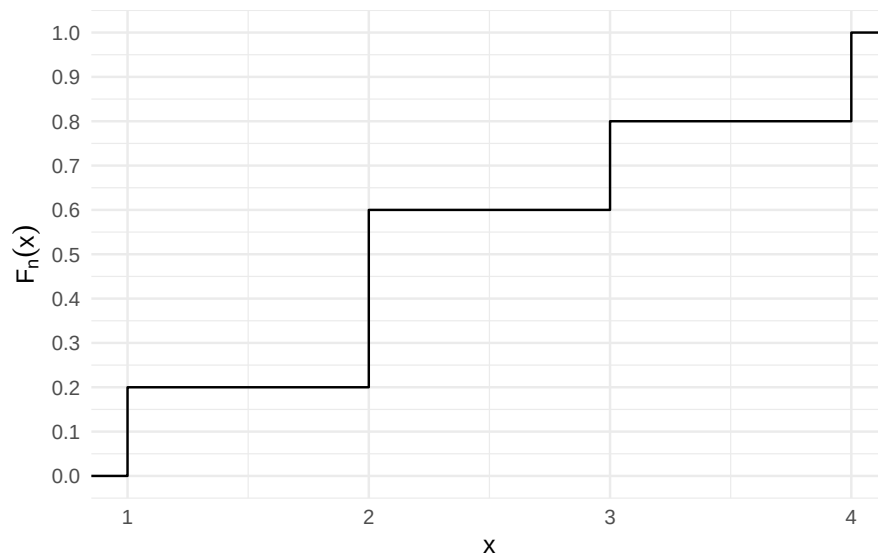


Homework problems:

Exercise 5.1. The figure below presents the empirical distribution function

$$F_n(x) := \frac{\text{number of elements in the sample } \leq x}{n} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{(-\infty, x]}(x_i)$$

for $n = 100$ observations $x_1, \dots, x_n$ of a numeric variable X .



- Identify the distinct values observed for the variable X .
- Calculate both the relative and absolute frequencies for each observed value of X , using the figure as a reference.
- Determine the empirical mean $\bar{x}_n$ and the median Q_2 .
- Suppose three additional observations, each with a value of 5, are drawn from the population. Analyze how the empirical mean and median change when calculated for all 103 observations compared to the previous values.

Solution:

- The observed values of the numeric variable are 1, 2, 3, 4.
- The figure shows the empirical distribution function, given by

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{(-\infty, x]}(x_i).$$

The empirical distribution function exhibits discontinuities at the observed variable values 1, 2, 3, and 4. Relative frequencies are obtained directly from the graph, while absolute frequencies are calculated by multiplying the relative frequencies by the total number of observations n .

Value	relative frequency	absolute frequency
1	0.2	20
2	0.4	40
3	0.2	20
4	0.2	20

- c) The figure is constructed from the observations $x_1, \dots, x_n$, with $n = 100$, and values 1,2,3,4. Therefore, we get for the mean

$$\begin{aligned}\bar{x}_n &= \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{100} (\underbrace{1 + \dots + 1}_{20 \text{ times}} + \underbrace{2 + \dots + 2}_{40 \text{ times}} + \underbrace{3 + \dots + 3}_{20 \text{ times}} + \underbrace{4 + \dots + 4}_{20 \text{ times}}) \\ &= \frac{1}{100} (20 \cdot 1 + 40 \cdot 2 + 20 \cdot 3 + 20 \cdot 4) = 2.4.\end{aligned}$$

Since the sample size is $n = 100$, an even number, the median is the average of the two central values in the ordered data set. According to the table in part b), both central values are 2. Thus,

$$Q2 = \frac{1}{2}(2 + 2) = 2$$

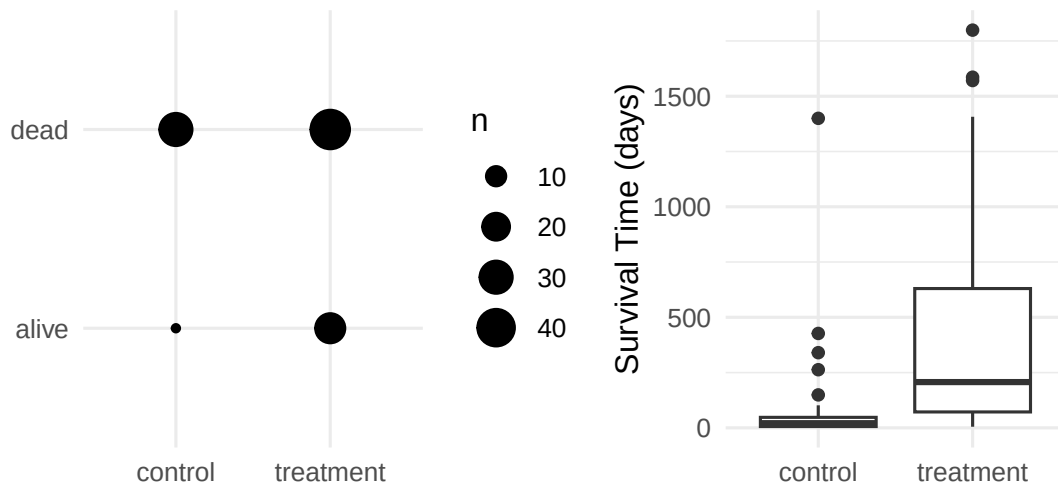
is the median value.

- d) To compute the new empirical mean, add the three additional values to the previous sum of observations and divide by 103:

$$\bar{x}_n = \frac{1}{103} (20 \cdot 1 + 40 \cdot 2 + 20 \cdot 3 + 20 \cdot 4 + 3 \cdot 5) = \frac{255}{103} \approx 2.48.$$

With the sample size now odd, the median corresponds to the middle value in the ordered data set. Since $x_{(52)}$ remains 2, the median is unchanged after adding the three new observations.

Exercise 5.2. The Stanford University Heart Transplant Study aimed to evaluate whether participation in an experimental heart transplant program increased patient lifespan. All individuals enrolled in the program were classified as official heart transplant candidates, indicating that they were critically ill and likely to benefit from transplantation. Patients were divided into two groups: the treatment group, who received a transplant, and the control group, who did not. The variable **transplant** identified group assignment. Among the 34 patients in the control group, 30 died, while 45 of the 69 patients in the treatment group died. The variable **survived** indicated whether each patient was alive at the conclusion of the study.



- Based on the count plot, is survival independent of whether or not the patient received a transplant? Explain your reasoning.
- What do the box plots reveal regarding the effectiveness of heart transplant treatment? Describe the shape of the distribution of survival times for both the treatment and control groups.
- What proportion of patients in the treatment group and in the control group did not survive?
- One approach to assess the effectiveness of the treatment is to apply a randomization technique. Using the method described at the end of Chapter 6, evaluate whether the data provide convincing evidence of an association between the variables **transplant** and **survived**.

Hint: To apply the randomization technique, raw data are required rather than absolute frequencies. Therefore, generate the data using the following code:

```
df <- tibble(
  transplant = c(rep('treatment', 69), rep('control', 34)),
  survived = c(rep(c('yes', 'no'), c(24, 45)), rep(c('yes', 'no'), c(4, 30)))
)
```

The code uses the **rep()** function to appropriately repeat the values of the variables **transplant** and **survived**. You can now apply the technique using the tibble **df**.

Solution:

- The proportion of patients alive at the end of the study is higher in the treatment group compared to the control group. These data suggest that survival is associated with receipt of a transplant.
- The median survival time for the control group is substantially lower than that of the treatment group. Recipients of a transplant generally exhibited longer survival. This finding is further supported by the substantially lower variability in the control group, as indicated by an interquartile range (IQR) of approximately 50 days compared to 500 days in the treatment group. Patients who did not receive a heart transplant tended to die much earlier and with less variability than those who received a transplant. Few patients in the control group survived beyond one year, whereas nearly half of the transplant recipients survived at least one year. The distribution of survival times in both groups is right-skewed. There is one clear outlier in the control group and additional potential outliers at the upper end in both groups.
- The proportion of patients in the treatment group who died is $\frac{45}{69} = 0.652$. The proportion of patients in the control group who died is $\frac{30}{34} = 0.882$.
- A randomization approach is employed to investigate whether there is evidence against the null hypothesis.

H_0 : The variables **transplant** and **survived** are independent. There is no significant relationship between the two groups, and the difference in survival rates between the control and treatment groups occurred by chance. In conclusion, a heart transplant is ineffective.

The alternative hypothesis is then given by:

H_A : The variables **transplant** and **survived** are not independent. The survival rates between the control and treatment groups significantly differed, indicating that the heart transplant procedure is effective.

The analysis begins by constructing the original dataset:

```
df <- tibble(
  transplant = c(rep('treatment', 69), rep('control', 34)),
  survived = c(rep(c('yes', 'no'), c(24, 45)),
               rep(c('yes', 'no'), c(4, 30)))
)
```

Next, the percentage of survival in the treatment group is recalculated.

```
treatment_per <- df |>
  filter(transplant == 'treatment', survived == 'yes') |>
  summarise(per_surv = n() / nrow(filter(df, transplant == 'treatment')))

treatment_per
```

```
# A tibble: 1 x 1
  per_surv
    <dbl>
1    0.348
```

In the control group the following value is obtained:

```
control_per <- df |>
  filter(transplant == 'control', survived == 'yes') |>
  summarise(per_surv = n() / nrow(filter(df, transplant == 'control')))

control_per
```

```
# A tibble: 1 x 1
  per_surv
    <dbl>
1    0.118
```

The difference between the two proportions is considered as the test statistic:

```
diff_stat <- treatment_per - control_per
diff_stat
```

```
per_surv
1 0.230179
```

Idea: Randomly permute the values of the variable **survived** while keeping the variable **transplant** unchanged. If the two variables are independent, the resulting statistic should be similar to the one observed in the original data. We need to repeat this process multiple times.

This procedure is repeated $N = 100$ times.

```
set.seed(12345) # running this command makes the result reproducible
N <- 100

diff_stat_perm <- vector("numeric", N)

for(i in 1:N){
  df_perm <- df |>
    mutate(survived = sample(survived, nrow(df)))
  df_perm_perc <- df_perm |>
    group_by(transplant, survived) |>
    summarise(n = n()) |>
    mutate(freq = n / sum(n))

  diff_stat_perm[i] <- df_perm_perc$freq[4] - df_perm_perc$freq[2]
}
```

The decision is now based on the relative frequency:

```
sum(abs(diff_stat_perm) >= diff_stat) / N
```

```
[1] 0
```

The relative frequency represents the proportion of simulated differences that are larger, in absolute value, than the observed difference in the data.

This result indicates that no simulated difference was as large, in absolute value, as 0.230179 in 100 simulation runs.

Therefore, the data provide compelling evidence for the alternative hypothesis of dependence between the variables **transplant** and **survived**. This result strongly supports the effectiveness of the transplant program.

Exercise 5.3 (5 points).

Grade all three submissions assigned to you (one being your own) in the peer-review workshop for Problem H.4.1 by May 18, 09:00.

You find the submissions by re-opening the peer-review workshop Problem H.4.1.

Exercise 5.4.



Complete the IDSST R tutorial **explore-num-data** tutorial by May 18.